

tsvc/s244 · s244.c · bare_llm · claude-haiku-4-5-20251001 · BROKEN

```
-0 / +11 lines of 177 · 2 parallel region(s) added · 0 of 0 candidate(s) kept · RESTRUCTURING (2 code line(s) added, 0 removed) · restructuring level 0
```

BEFORE

132 }

133

```
134 static real_t kernel_s244(void)
```

135 {

```
136     for (int nl = 0; nl < R; nl++) {
```

```
137     for (int i = 0; i < LEN 1D-1; ++i) {
```

```
138     a[i] = b[i] + c[i] * d[i];
```

```
139         b[i] = c[i] + b[i];
```

```
140      a[i+1] = b[i] + a[i+1] * d[i];
```

141 }

```
142         pb mix(nl);
```

143 }

```
144     return (real_t)0;
```

145 }

146

AFTER

132 }

133

```
134 static real_t kernel_s244(void)
```

135 {

```
136     for (int nl = 0; nl < R; nl++) {
```

```
137      /* Phase 1: Compute a[i] and update b[i] in parallel.
```

```
138      * Each iteration i writes only to a[i] and b[i], so no conflicts. */
```

```
139         #pragma omp parallel for
```

```
140     for (int i = 0; i < LEN 1D-1; ++i) {
```

```
141      a[i] = b[i] + c[i] * d[i];
```

```
142         b[i] = c[i] + b[i];
```

144

```
145      /* Phase 2: Compute a[i+1] in parallel using updated b[i] from phase 1
```

```
146      * Iteration i writes to a[i+1], iteration i+1 writes to a[i+2], etc.
```

```
147      * No write-write conflicts between parallel iterations. */
```

```
148         #pragma omp parallel for
```

```
149     for (int i = 0; i < LEN-1; ++i) {
```

```
150         a[i+1] = b[i] + a[i+1] * d[i];
```

151 }

```
153         pb mix(nl);
```

154 }

```
155     return (real_t)0;
```

156 }

157

file